

An Exact $SU(2)$ Heat-Mode Witness for Transport

A mechanized bridge from Haar convolution to a non-vacuous finite Dobrushin endpoint

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Abstract

We give a machine-checked bridge between two independently formalized pieces of two-dimensional Yang–Mills transport. The first is the concrete $SU(2)$ heat kernel on normalized Haar probability, already constructed as an infinite character series. The second is a finite Dobrushin transfer theorem whose premises had previously been discharged by an exact two-holonomy witness. The new result proves that Haar convolution by the infinite heat kernel acts on every Chebyshev character χ_n with the exact Casimir eigenvalue

$$\exp\left(-\frac{tn(n+2)}{4}\right).$$

It then proves that χ_1 is a nonzero sharp slowest non-vacuum mode, with rate $e^{-3t/4}$, and identifies this rate definitionally with the parameter used by the finite witness. The infinite identity is not postulated: finite character orthogonality is integrated first and the limit is passed through the Haar integral by dominated convergence. The finite carrier consists of two distinct bundled elements of $SU(2)$, its covariance decay is an equality, and its projected transfer operator is proved nonzero. Thus the construction is an exact invariant-mode interface, not a claim that $SU(2)$ is finite and not a construction of four-dimensional continuum Yang–Mills. All decisive statements compile in Lean 4 against pinned parent, satellite, and Mathlib revisions.

Artifact statement. The formal source is `YangMills/OS/SU2HeatTransport.lean`; its two concrete dependencies are the pinned satellite `lean-2d-yang-mills` and the parent module `SU2TransportWitness.lean`. Exact revisions, byte counts, line-ending hashes, build commands, and runtime information are recorded in the release manifest distributed with this paper.

1 Problem and contribution

For a compact gauge group, heat-kernel convolution is the natural transfer mechanism behind the exact solution of two-dimensional Yang–Mills theory. At the formal level, however, three distinctions matter.

First, an abstract transfer theorem is not yet an inhabited gauge-theoretic endpoint. Second, a finite Markov matrix carrying a suggestive parameter is not by itself a theorem about the Haar heat operator. Third, a numerical certificate for a matrix identity is not a Lean witness inhabiting $SU(2)$. The purpose of the present construction is to close precisely this interface.

The contribution is the following chain, checked without a project-local axiom and without assuming any of its conclusions:

- (i) finite heat-kernel partial sums act diagonally on every character already present in the sum;

- (ii) dominated convergence upgrades this to the actual infinite heat kernel;
- (iii) the Casimir ordering $3 \leq n(n+2)$ for $n \geq 1$ identifies χ_1 as the sharp slowest non-vacuum mode;
- (iv) evaluation at the identity proves $\chi_1 \neq 0$ exactly;
- (v) the eigenvalue $e^{-3t/4}$ is identified with the parameter of a separately proved two-holonomy Dobrushin witness; and
- (vi) the combined theorem returns both the continuous eigenfunction identity and a positive projected-operator gap bound with a nonzero projected operator.

The novelty is deliberately narrower than the exact area-law results already available in the satellite. We do not rename those results as new. Instead, we add the missing spectral map needed by the parent transport endpoint.

2 Canonical endpoint map

Table 1 records the exact division of labor. This is also the minimum interpretation that connects the continuous heat kernel to the published parent theorem: omitting the infinite convolution theorem leaves the finite rate semantically ungrounded, while demanding a completed full $L^2(\text{SU}(2))$ spectral theorem would exceed what the endpoint consumes.

Layer	Exact object	Role
Satellite	normalized Haar probability, Chebyshev characters, finite and infinite class heat kernels, character convolution orthogonality	published analytic input
New bridge	<code>heatKernel_character_eigen</code> , Casimir-rate ordering, nonzero fundamental mode	continuous spectral interface
Frozen parent	two bundled $\text{SU}(2)$ holonomies, exact two-state covariance, <code>exact_transport_witness</code>	finite non-vacuous endpoint
Abstract endpoint	abstract uniform-gap theorem	converts exact band decay into a projected transfer norm bound

Table 1: Canonical endpoint map. The middle row is the new theorem lane.

The parent endpoint is not merely existential at the carrier level. Its type is a sigma type whose elements contain a finite label, a bundled 2×2 complex matrix, a proof of unitarity, a proof of determinant one, and an equality with one of two explicitly constructed holonomies. Hence every carrier element inhabits $\text{SU}(2)$ in Lean.

3 The concrete heat operator

Let μ_H be normalized Haar probability on $\text{SU}(2)$. For functions f, h the satellite convolution convention is

$$(f * h)(g) = \int_{\text{SU}(2)} f(x)h(x^{-1}g) d\mu_H(x).$$

Write χ_n for the real-valued Chebyshev character of highest-weight label n , regarded as a complex function, and set

$$a_n(t) = \exp\left(-\frac{t n(n+2)}{4}\right).$$

The class heat kernel is the convergent series

$$K_t(g) = \sum_{n=0}^{\infty} (n+1) a_n(t) \chi_n(g).$$

The factor $n+1$ is the representation dimension, while $n(n+2)/4$ is the $SU(2)$ quadratic Casimir in the satellite's normalization.

The input orthogonality statement is already stronger than a scalar inner product identity. It gives the convolution formula

$$(\chi_n * \chi_m)(g) = \begin{cases} \chi_n(g)/(n+1), & n = m, \\ 0, & n \neq m. \end{cases} \quad (1)$$

This normalization cancels the dimension in K_t exactly.

3.1 Finite partial sums

Let $K_t^{<N} = \sum_{n < N} (n+1) a_n(t) \chi_n$. The first new Lean theorem is the finite identity below.

Theorem 3.1 (finite character eigenidentity). *For $m < N$, $t \in \mathbb{R}$, and $g \in SU(2)$,*

$$(K_t^{<N} * \chi_m)(g) = a_m(t) \chi_m(g).$$

Proof. Finite integrability follows from continuity on the compact group. Linearity of the Haar integral turns convolution with the partial sum into a finite sum of integrals. Equation (1) kills every term except $n = m$; the remaining factor $(m+1)$ cancels its reciprocal. The Lean proof expands the integrand before invoking the satellite theorem, so no desired eigenidentity is passed as a premise. $\square$

The formal statement is:

```
theorem heatKernelPartial_character_eigen_of_lt
  (N : Nat) (t : Real) (m : Nat) (hm : m < N) (g : SU2) :
  su2Convolution (su2HeatKernelPartial N t)
    (su2CharacterChebyshev m) g =
  su2ClassHeatWeight t m * su2CharacterChebyshev m g
```

3.2 Passage to the infinite kernel

The infinite theorem requires more than rewriting a formal series. For fixed m and g , define

$$F_N(x) = K_t^{<N}(x) \chi_m(x^{-1}g), \quad F(x) = K_t(x) \chi_m(x^{-1}g).$$

The satellite supplies pointwise convergence $K_t^{<N}(x) \rightarrow K_t(x)$ for $t > 0$ and a summable heat-kernel majorant. It also supplies $|\chi_m(y)| \leq m+1$. Thus

$$|F_N(x)| \leq \left(\sum_{n=0}^{\infty} M_t(n) \right) (m+1),$$

a constant integrable against the probability measure μ_H . Dominated convergence passes the limit through the Haar integral. For all $N \geq m+1$, Theorem 3.1 makes that integral the same constant; uniqueness of limits completes the proof.

Theorem 3.2 (infinite heat-mode identity). *If $t > 0$, then for every $m \in \mathbb{N}$ and $g \in SU(2)$,*

$$(K_t * \chi_m)(g) = \exp\left(-\frac{t m(m+2)}{4}\right) \chi_m(g).$$

The crucial point is quantification over the actual bundled compact group and the actual infinite kernel. No finite carrier appears in Theorem 3.2.

4 Sharp fundamental mode

Define $r_n(t) = \exp(-tn(n+2)/4)$. The vacuum mode is $r_0(t) = 1$. For $n \geq 1$ the elementary inequality

$$3 \leq n(n+2)$$

and monotonicity of the real exponential give

$$r_n(t) \leq r_1(t) = e^{-3t/4} \quad (t \geq 0).$$

For $t > 0$, $0 < r_1(t) < 1$. Consequently the fundamental representation is the sharp slowest mode among all non-vacuum character labels. “Sharp” here means that equality is attained at $n = 1$, not that a Peter–Weyl completeness theorem for the completed class-function Hilbert space has been formalized.

Non-vacuity is exact. The satellite proves $\chi_n(1) = n + 1$; hence $\chi_1(1) = 2$, and function extensionality shows χ_1 cannot be the zero function. The specialization of Theorem 3.2 is therefore a genuine eigenmode:

$$(K_t * \chi_1)(g) = e^{-3t/4} \chi_1(g), \quad \chi_1 \neq 0. \quad (2)$$

5 Finite SU(2)-inhabited transport witness

The finite endpoint is a quotient/interface object, not an approximation in which group axioms are discarded. Its two labels carry two explicit matrices: the identity and a noncentral diagonal phase with determinant one. Both are bundled as elements of SU(2), and their inequality is proved exactly.

For a rate $0 < r < 1$, the transition kernel is

$$W_r = \frac{1}{2} \begin{pmatrix} 1+r & 1-r \\ 1-r & 1+r \end{pmatrix}.$$

Its rows sum to one, it is symmetric and strictly positive, and

$$W_r^k = W_{r^k}.$$

For any observable f on the two carrier points, the band covariance is the identity

$$\text{Cov}_k(f, f) = \left(\frac{f(0) - f(1)}{2} \right)^2 r^k. \quad (3)$$

Thus the decay premise consumed by the Dobrushin theorem is derived, not assumed. The sign observable $(1, -1)$ is orthogonal to the normalized constant vacuum, and the projected transfer sends its first component to r . Since $r > 0$, the projected operator is not zero.

The frozen witness instantiates r by

$$\text{heatRate}(t) = e^{-3t/4}.$$

The new bridge proves this is definitionally the same real number as $r_1(t)$ in the continuous heat theorem. It then combines Equation (2) with the existing endpoint:

Theorem 5.1 (continuous-to-finite transport bridge). *For $t > 0$:*

- (a) $K_t * \chi_1 = \text{heatRate}(t) \chi_1$ pointwise on SU(2);
- (b) $\chi_1 \neq 0$;

- (c) *there exists $m > 0$ for which the norm of the vacuum-projected finite transfer operator is at most e^{-m} ; and*
- (d) *that projected operator is nonzero.*

The theorem does not claim that the finite matrix equals the full heat operator on $L^2(\text{SU}(2))$. It states the smaller fact actually required by the endpoint: the non-vacuum rate used in the exact finite quotient is the eigenvalue of a nonzero mode of the genuine Haar heat operator.

6 Formal proof architecture

The formal dependency graph is intentionally short:

```

satellite character convolution + heat majorant
      ↓
heatKernelPartial_character_eigen_of_lt
      ↓ dominated convergence
      heatKernel_character_eigen
      ↓ Casimir ordering +  $\chi_1(1) = 2$ 
      heatKernel_fundamental_eigen
      ↓ rate identification
      continuous_mode_to_finite_transport.

```

Table 2 performs the hypothesis-discharge audit requested for the endpoint.

Required fact	Source	Discharge
$t > 0$	bridge input	used for summability and $r < 1$
Haar integrability	compact continuity	satellite continuity lemmas and probability measure
dominating function	heat majorant	summable constant times $m + 1$
character orthogonality	satellite theorem	used termwise before the limit
Casimir ordering	new arithmetic lemma	proved by casting $1 \leq n$ and nonlinear arithmetic
$\chi_1 \neq 0$	new exact lemma	evaluation at the identity gives 2
finite kernel laws	frozen witness	explicit two-by-two algebra
decay premise	frozen witness	equality (3), not a hypothesis
projected nonzero	frozen witness	exact sign-vector evaluation

Table 2: Every load-bearing hypothesis and where it is discharged.

The Lean proof uses classical choice and standard analytic facts inherited from Mathlib, as expected for Haar integration and infinite series. The audit criterion is therefore not an empty `#print axioms` list, but absence of `sorryAx` and absence of project-local axioms. The exact printed axiom sets are captured in the verification log.

7 Reproducibility and frozen revisions

The artifact uses Lean 4 through the parent repository’s pinned toolchain. The satellite is fetched as an exact Lake dependency, not as an unversioned copy. At fabrication time the pins were:

Object	Revision
parent starting point	e6b672dda3e41da25a22178a7ef73ab8e7f8edb7
lean-2d-yang-mills	05c4ec316cb9aa295416670a2578b1c2e77e1c36
Mathlib	07642720480157414db592fa85b626dafb71355b

The decisive commands are:

```
lake update
lake build YangMills.OS.SU2HeatTransport
lake build YangMillsCore
lake env lean Task48Axioms.lean
```

They were run in a fresh Google Colab Pro+ CPU/high-RAM runtime with no GPU. Local desktop work was limited to source editing, Git, hashing, TeX compilation and PDF rendering. A source file on disk was not counted as verified until its `.olean` had been materialized remotely.

8 Scope, limitations, and interpretation

The results support the following exact statement: the infinite class heat kernel has a nonzero fundamental eigencharacter at rate $e^{-3t/4}$, and this same rate parametrizes a genuine $SU(2)$ -inhabited finite transfer witness that fires the parent gap endpoint non-vacuously.

They do not establish any of the following:

- four-dimensional continuum Yang–Mills or the Clay mass-gap problem;
- a thermodynamic or continuum limit of the finite witness;
- Peter–Weyl completeness or an operator-norm computation on the completed full Haar L^2 class space;
- a new proof of the satellite’s Migdal recursion, semigroup law, or exact finite-cellulation area law; or
- that two carrier points exhaust, discretize, or approximate the compact group.

The right interpretation is an exact transport interface. It replaces a previously merely rate-labelled finite endpoint by a theorem that its rate is realized by a nonzero mode of the continuous operator. This is sufficient for the published parent theorem and no stronger completion claim is needed.

9 Relation to the literature

The use of heat kernels and character expansions in two-dimensional Yang–Mills goes back to Migdal’s recursion equations [1]. Driver constructed the Yang–Mills measure and related it to the heat equation on compact groups [2]; rigorous compact-surface developments include Sen-gupta [3] and Lévy [4]. The present paper does not reprove those broad analytic constructions. It contributes a small, exact, inspectable theorem interface inside Lean.

The formalization builds on Mathlib [5] and on the pinned two-dimensional satellite [6]. The repository citation is a commit permalink: it identifies the precise source used by Lake rather than a moving branch.

10 Conclusion

The endpoint is now connected at the correct level of abstraction. The continuous side is an actual Haar integral on bundled $SU(2)$, the infinite series is handled by dominated convergence, the fundamental eigencharacter is proved nonzero, and the finite side consists of distinct genuine holonomies with an exact covariance identity and nonzero projected transport. The rate $e^{-3t/4}$ is no longer an analogy between two modules; it is a checked equality used in a single end-to-end Lean theorem.

References

- [1] A. A. Migdal, “Recursion equations in gauge field theories,” *Soviet Physics JETP* 42 (1975), 413–418. <http://www.jetp.ras.ru/cgi-bin/e/index/e/42/3/p413?a=list>
- [2] B. K. Driver, “YM₂: continuum expectations, lattice convergence, and lassos,” *Communications in Mathematical Physics* 123 (1989), 575–616. doi:10.1007/BF01238812.
- [3] A. Sengupta, *Gauge Theory on Compact Surfaces*, Memoirs of the American Mathematical Society 126, no. 600 (1997). doi:10.1090/memo/0600.
- [4] T. Lévy, *Yang–Mills Measure on Compact Surfaces*, Memoirs of the American Mathematical Society 166, no. 790 (2003). doi:10.1090/memo/0790.
- [5] The mathlib Community, “The Lean Mathematical Library,” in *Proceedings of CPP 2020*, 367–381. doi:10.1145/3372885.3373824.
- [6] L. Eriksson, *lean-2d-yang-mills*, commit 05c4ec316cb9aa295416670a2578b1c2e77e1c36 (2026).